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Course Number:	BME 705
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Instructor:	Dr. Somayeh Norouzi
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<i>Assignment/Lab Number:</i>	3
<i>Assignment/Lab Title:</i>	Lab 3: Center of pressure (COP) and sway analysis

<i>Submission Date:</i>	November 12th, 2025
<i>Due Date:</i>	November 12th, 2025

Student LAST Name	Student FIRST Name	Student Number	Section	Signature
Petrova	Irina	501126967	3	I. Petrova
Hassan	Tanvir	501104056	3	Tanvir Hassan

*By signing above you attest that you have contributed to this written lab report and confirm that all work you have contributed to this lab report is your own work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a "0" on the work, an "F" in the course, or possibly more severe penalties, as well as a Disciplinary Notice on your academic record under the Student Code of Academic Conduct, which can be found online at: <http://www.ryerson.ca/senate/current/pol60.pdf>

Background

Postural control is a fundamental aspect of human movement that allows people to maintain their balance during daily tasks. Balance is achieved through the integration of sensor information from visual, vestibular and muscular systems. They guide the motor responses to maintain the center of mass over the base of support [1]. The centre of pressure (COP) is a point on a surface where a single force can represent the total effect of the pressure acting on that surface. It is determined by the location of the COP relative to the object's center of gravity or base of support. Analyzing the COP in both medial-lateral (ML) and anterior-posterior (AP) directions provides insight into the dynamics of the postural control system [2].

Muscles provide strength and force needed to control posture and correct imbalances, and work in coordinated groups to stabilize the body against gravity and other environmental stimuli. This control relies on the central nervous system coordinating sensory input with motor responses. There are a few muscle groups that directly contribute to balance to maintain control such that :

- ➔ Postural control - provided by certain muscles called postural muscles like the core muscles and some specific back muscles.
- ➔ Corrective action - when the body is perturbed such as a slight swing during standing the specific leg and core muscles activate to generate forces to bring the body's center of mass back to its base.
- ➔ Muscle strength - strong muscles in the lower body and core are essential for responding quickly to balance challenges [4].

Additionally, the functional electrical stimulation (FES) can be used to activate muscles in response to losing balance. It can improve balance by strengthening muscles, enhancing muscle coordination during movement, and compensating for muscle paralysis or weakness. It works by sending electric impulses to activate muscles, which can improve static and dynamic balance, especially when combined with rehabilitative exercises [6].

Electromyography (EMG) signals, combined with COP measurements can provide valuable information on muscle activities and patterns during balance tasks [3].

Introduction

This lab focuses on exploring and evaluating the human balance using COP measurements from split-force plates. Data is recorded of participants performing quiet standing tasks with eyes open (EO) and eyes closed (EC), as well as slow and fast anterior-posterior sway.

The force plate data is used to compute the COP for the left (COPleft) and right (COPright) plates and then combined to determine the overall COP (COPnet). The overall COP has two components: x (medial-lateral, ML) and y (posterior-anterior, AP). Metrics such as total excursion (TOTEX) and mean velocity (MVELO) are calculated to quantify the sway and assess balance under different conditions [3].

Eyes open (EO) and eyes closed (EC) are compared as vision contributes to postural control. The balance system relies on three main inputs - visual, vestibular and somatosensory. When eyes are open the brain uses visual cues to determine the body's position in space. When eyes are closed, visual information is removed allowing the body to rely on more vestibular input to maintain balance. Comparing EO and EC conditions allows for understanding of how much vision helps stabilize posture. Typically, posture sway increases when eyes are closed in AP and ML directions. The difference is quantified using COP metrics like TOTEX and MVELO. This comparison can be useful in research and rehabilitation such as balance disorders, vestibular dysfunctions, or recovery after stroke [7].

Some applications of the sway-balance analysis include concussion assessment and management, rehabilitation, injury and risk evaluation and sports performance enhancement. Balance analysis is used to diagnose injuries and determine when athletes are able to safely return to play, comparing post and pre-injury scores. It also helps identify deficits to guide the development of personalized rehabilitation strategies. It can be used to evaluate risk of falls, especially in the elderly [5].

This lab experiment also incorporates the use of EMG analysis of the soleus and tibialis anterior muscles. This provides insight into muscle activation patterns during postural adjustments. MATLAB is used to analyze the data, including applying the provided functions to convert raw ground reaction forces into COP measurements [3].

Results & Figures

Part 1

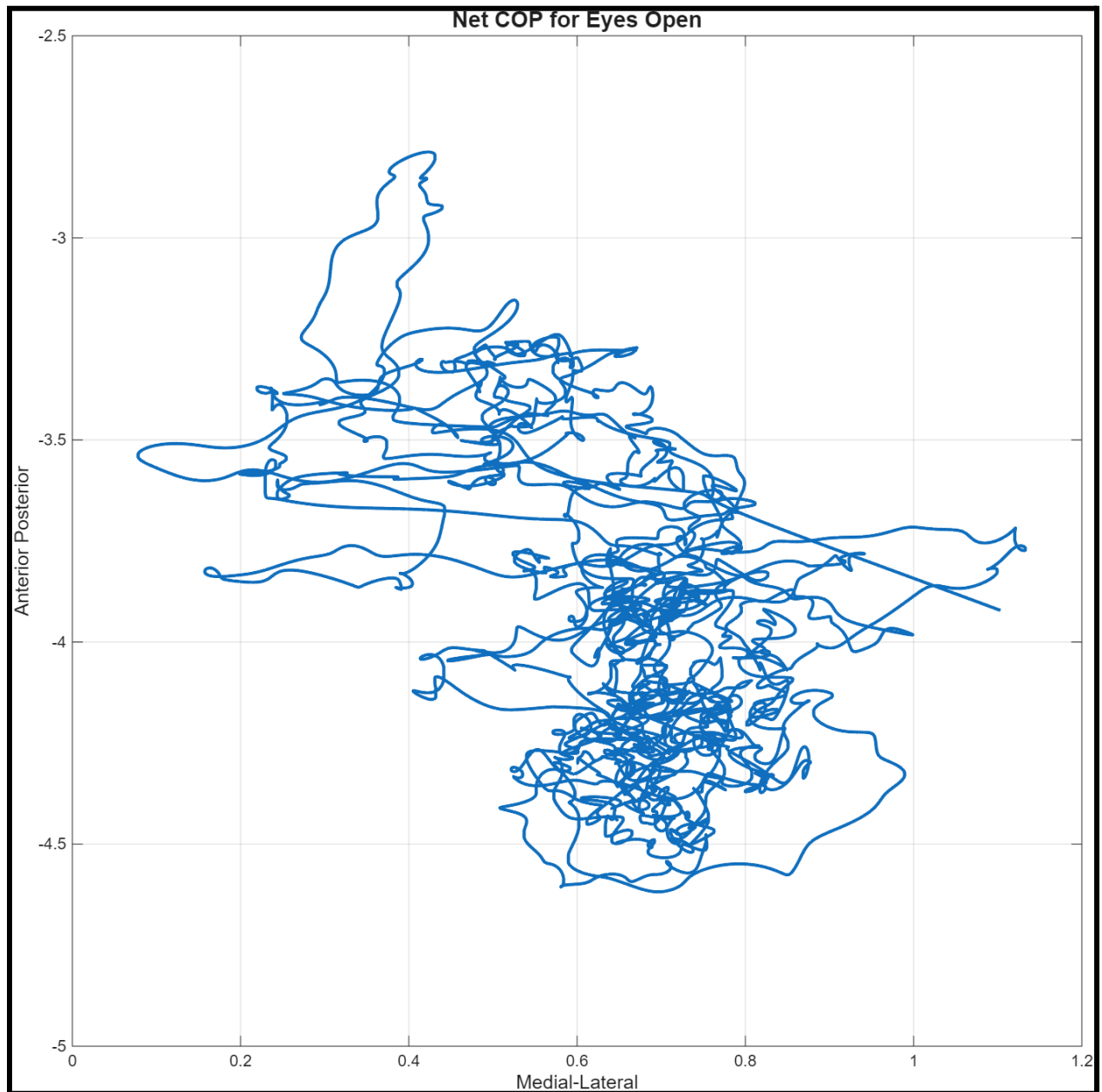


Figure 1: Stabilogram for Eyes Open [A.1]

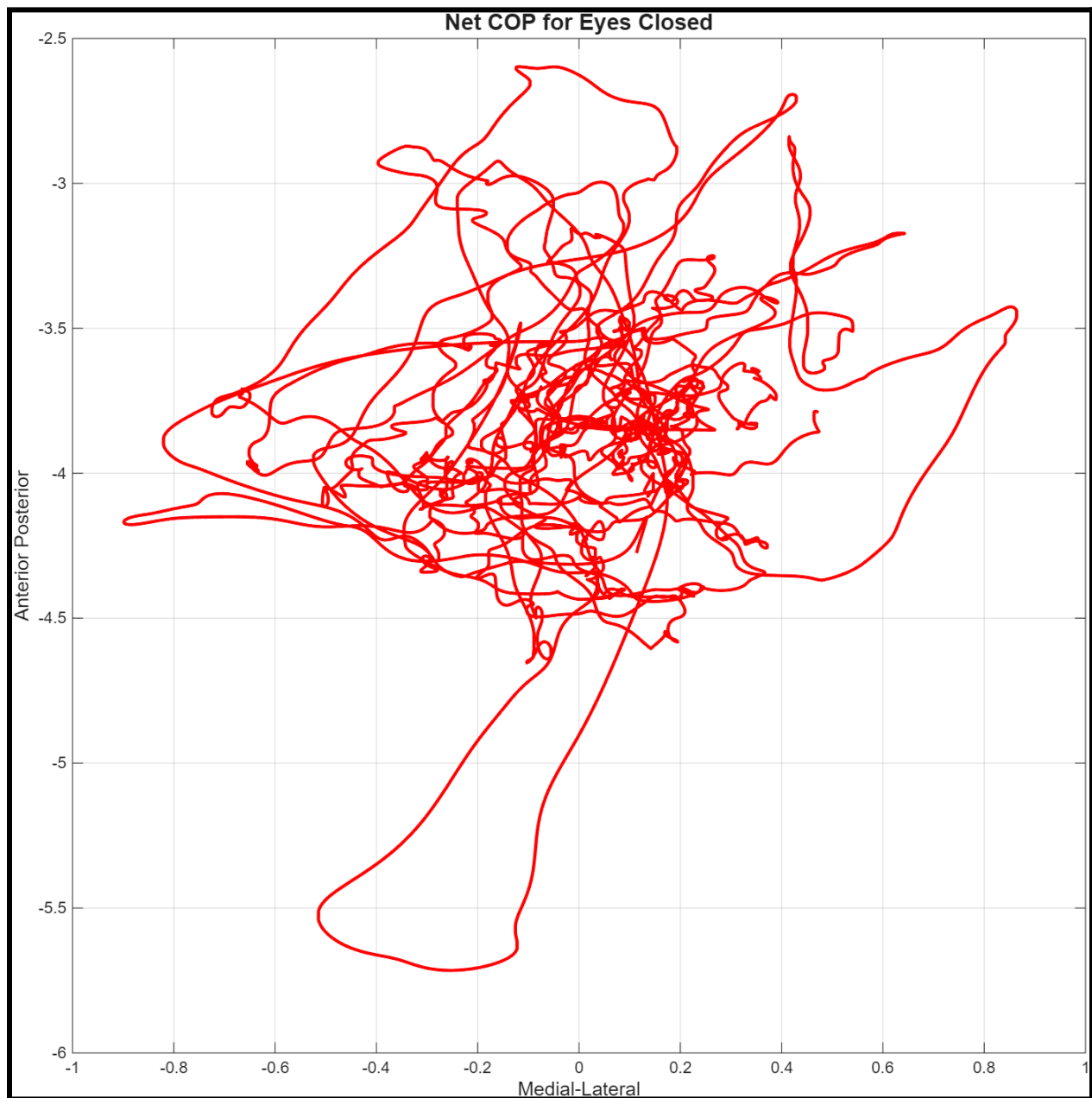


Figure 2: Stabilogram for Eyes Closed [A.1]

1.2: Compare EO and EC conditions and discuss the differences.

Comparing the stabilograms in Figures 1 and 2 highlights differences in postural control between the eyes open (EO) and eyes closed (EC) conditions. In Figure 1 (EO), the center of pressure (COP) trajectory is more compact, reflecting greater stability and reduced sway around the center. In contrast, Figure 2 (EC) displays larger deviations in COP, with the path extending farther from the central region, particularly along the anterior-posterior axis. This wider distribution indicates that, in the absence of visual input, the subject had more difficulty maintaining a stable COP position, resulting in increased sway and reduced postural control. These

observations agree with established findings that visual feedback is essential for sustaining balance and stability.

Metrics	EyesOpen	EyesClosed
{'Total Excursion (AP) (cm)' }	7.1619	8.8899
{'Total Excursion (ML) (cm)' }	9.6363	12.404
{'Total Excursion (Net) (cm)'} }	13.182	16.632
{'Mean Velocity (AP) (cm/s)' }	0.35809	0.44449
{'Mean Velocity (ML) (cm/s)' }	0.48181	0.62022
{'Mean Velocity (Net) (cm/s)'} }	0.65909	0.83162

Figure 3: Excursion & Velocity outputs.

Table 1: Total Excursion and Mean Velocities for AP, ML and Net [A.1].

Metrics	Eyes Open	Eyes Closed
Total Excursion (AP) (cm)	7.1619	8.8899
Total Excursion (ML) (cm)	9.6363	12.404
Total Excursion (Net) (cm)	13.182	16.632
Mean Velocity (AP) (cm/s)	0.35809	0.44449
Mean Velocity (ML) (cm/s)	0.48181	0.62022
Mean Velocity (Net) (cm/s)	0.65909	0.83162

1.4: Comment on the differences between excursions and velocities when the subject has their eyes closed vs eyes open

The numerical results presented in Table 1 support the observations discussed in Part 1.2 and provide quantitative evidence of the influence of visual input on postural stability. Under the eyes closed (EC) condition, total excursions in both the anterior-posterior (AP) and medial-lateral (ML) directions were greater than in the eyes open (EO) condition, with AP increasing from

7.16 cm to 8.89 cm and ML from 9.64 cm to 12.40 cm. The corresponding rise in net excursion, from 13.18 cm to 16.63 cm, reflects increased overall sway when visual input was removed.

Mean velocities followed a similar trend, with AP velocity increasing from 0.36 cm/s to 0.44 cm/s, ML from 0.48 cm/s to 0.62 cm/s, and net mean velocity from 0.66 cm/s to 0.83 cm/s. These higher velocities indicate that, without visual feedback, the center of pressure (COP) movements became both larger and faster. These results align with the patterns observed in the stabilograms from Part 1.2, reinforcing that visual information plays a key role in minimizing sway amplitude and velocity, thereby enhancing balance control.

Part 2

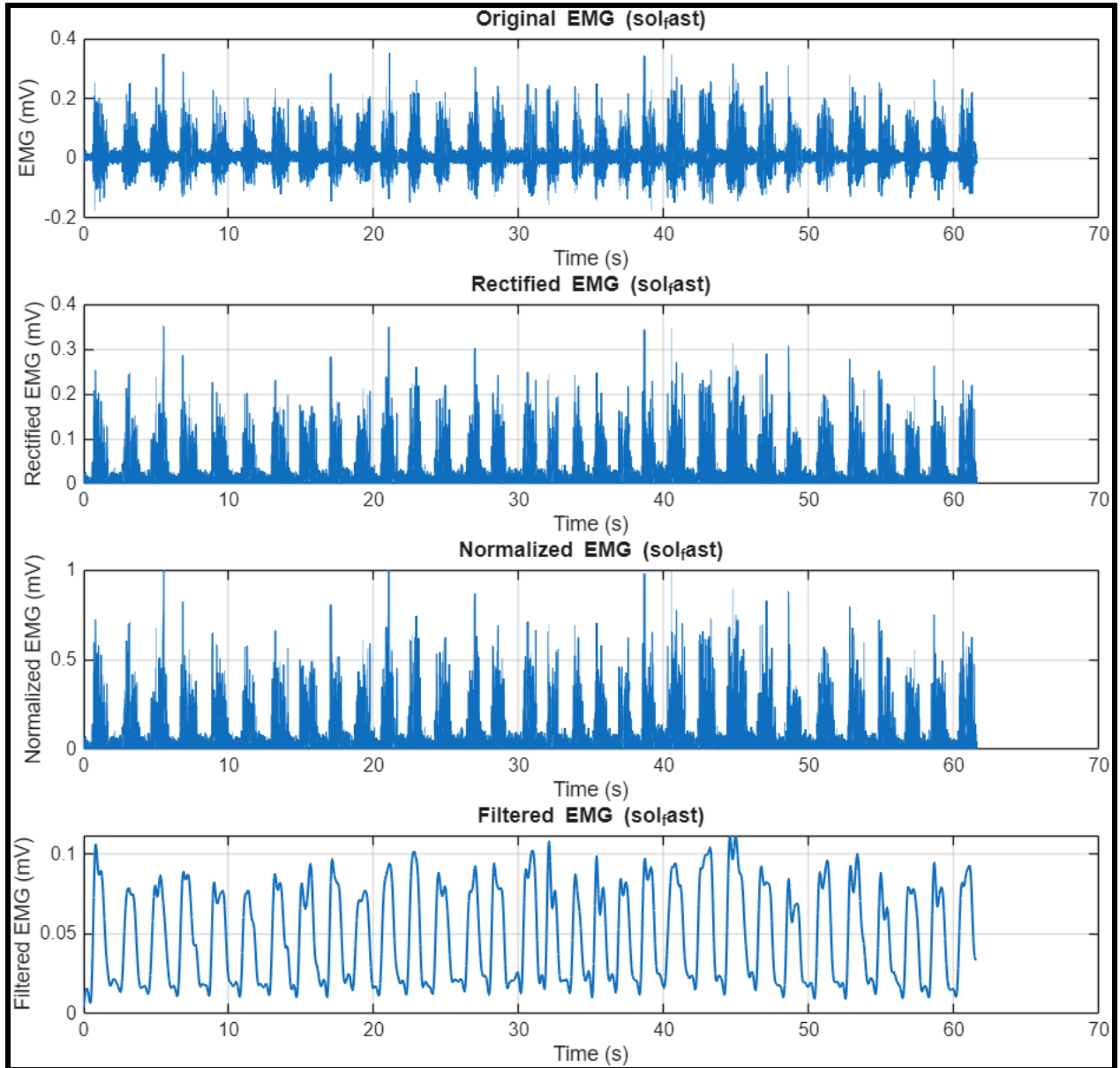


Figure 4: Processing of the sol fast sway.

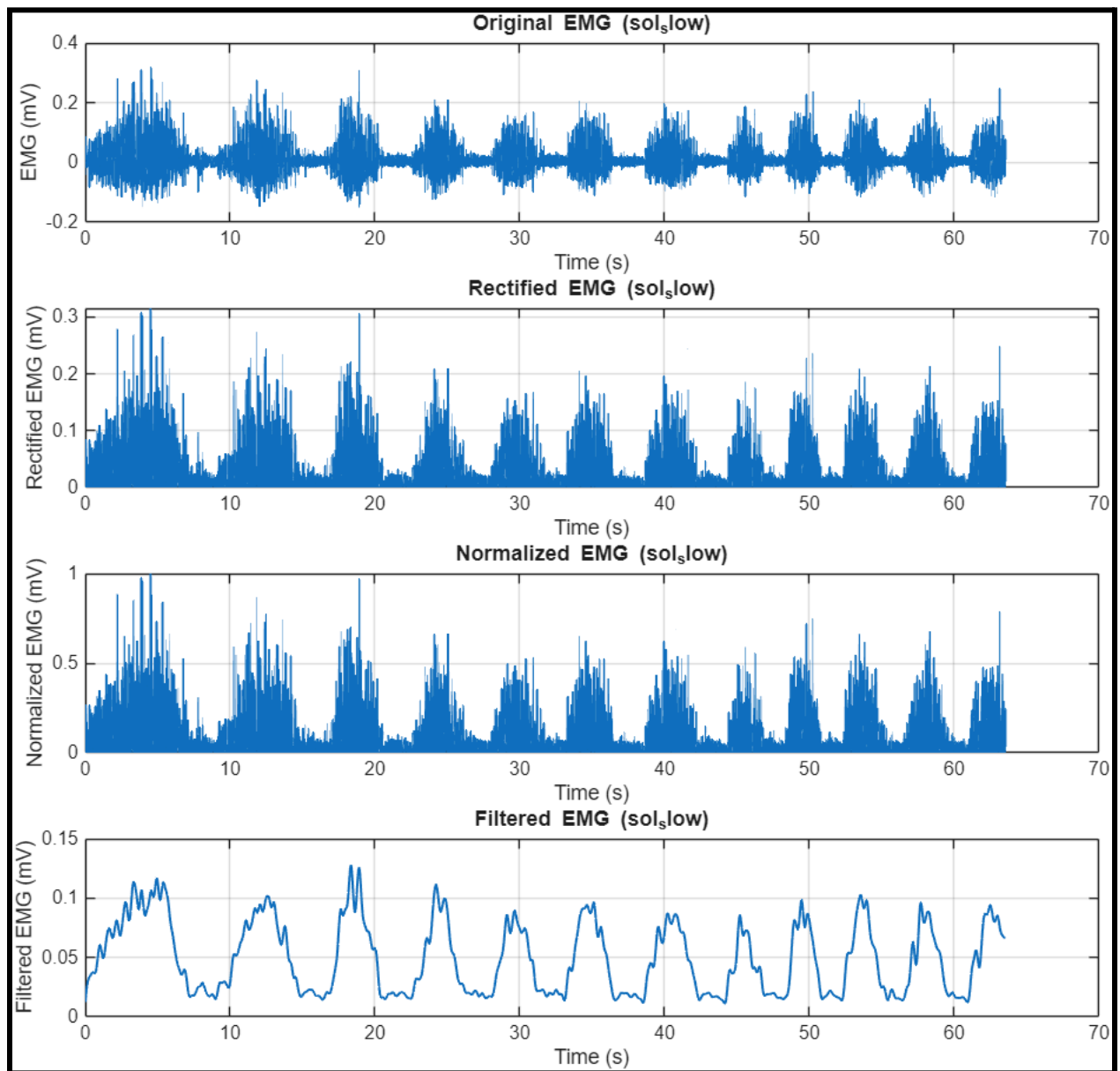


Figure 5: Processing of the sol slow sway.

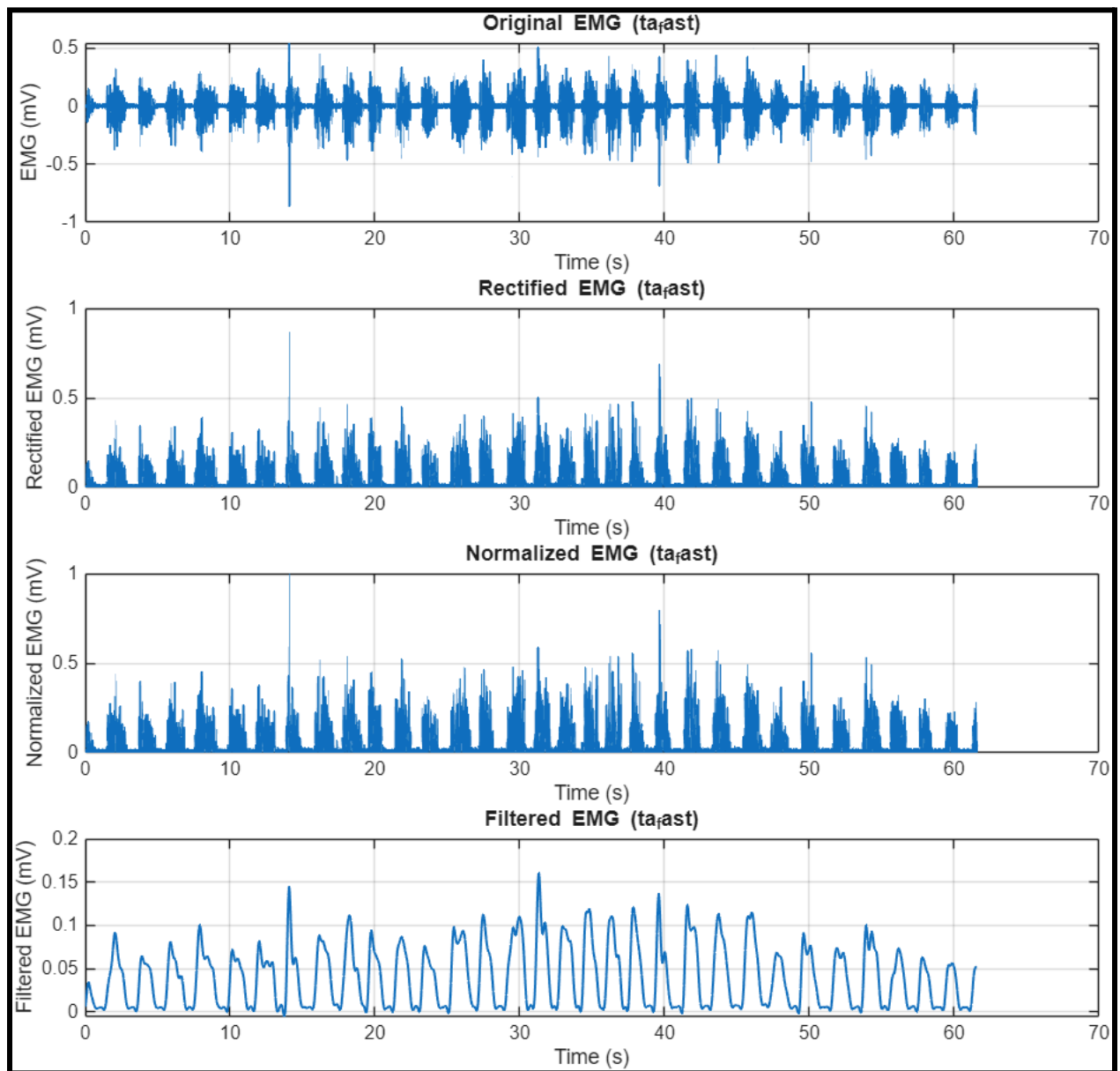


Figure 6: Processing of the ta fast sway.

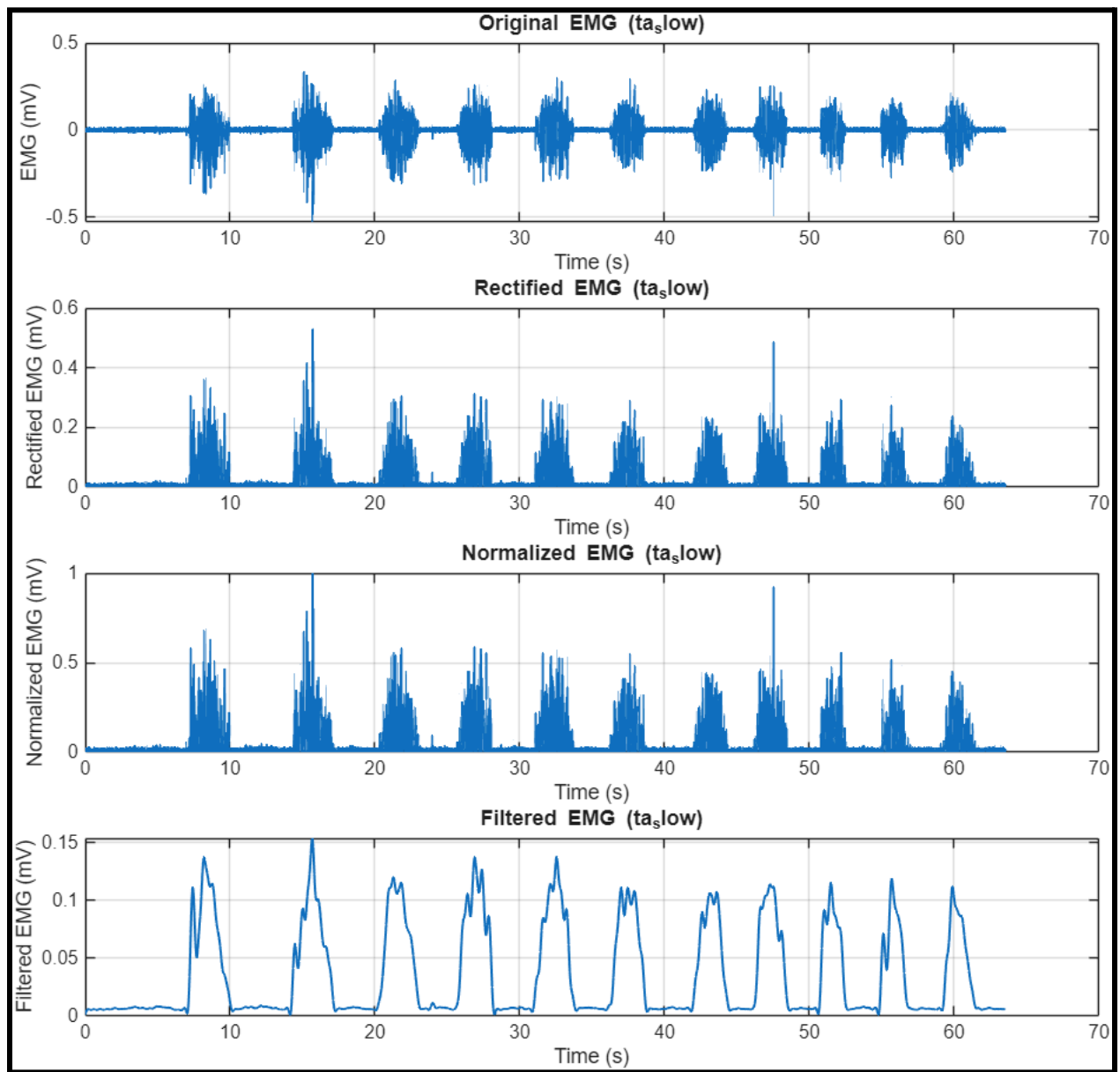


Figure 7: Processing of the ta slow sway.

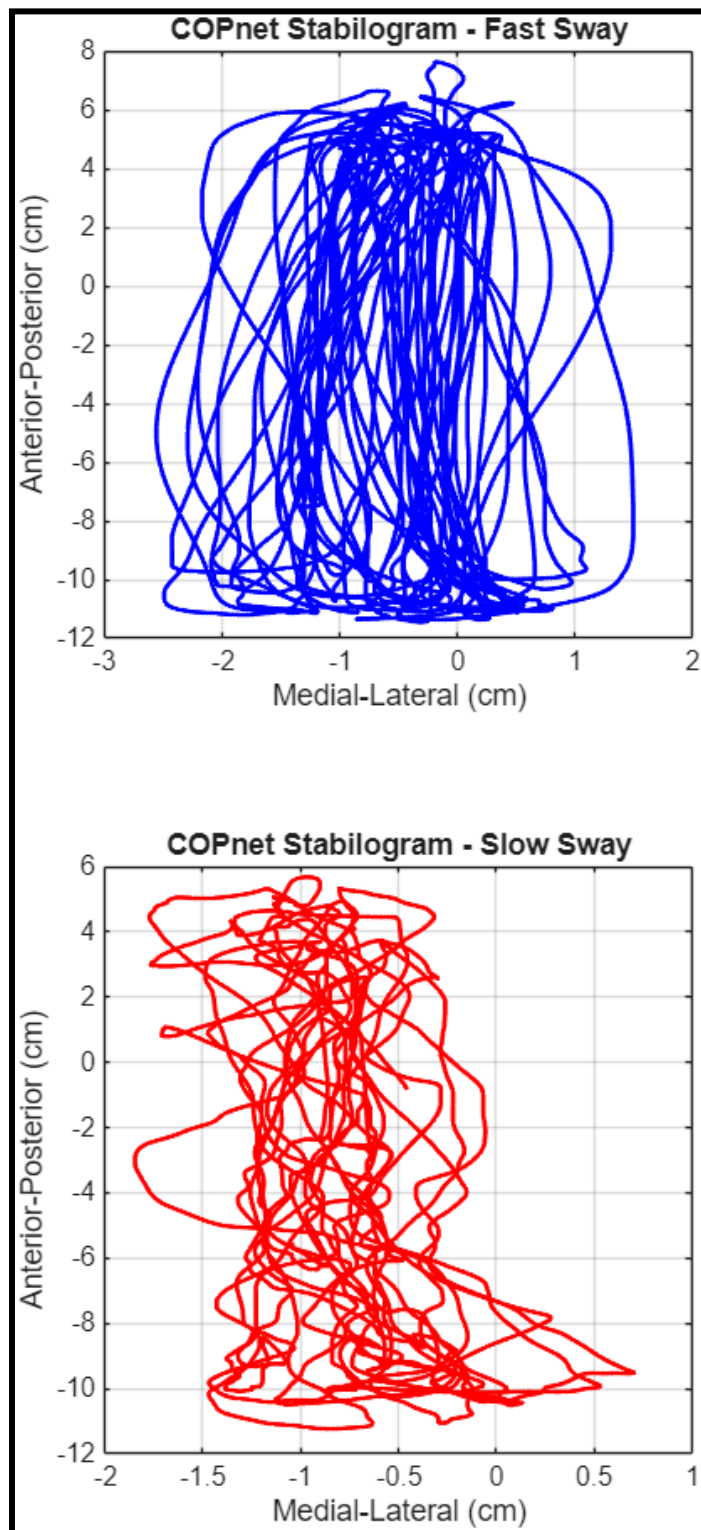


Figure 8: Fast and Slow Stabilograms for COP [A.2]

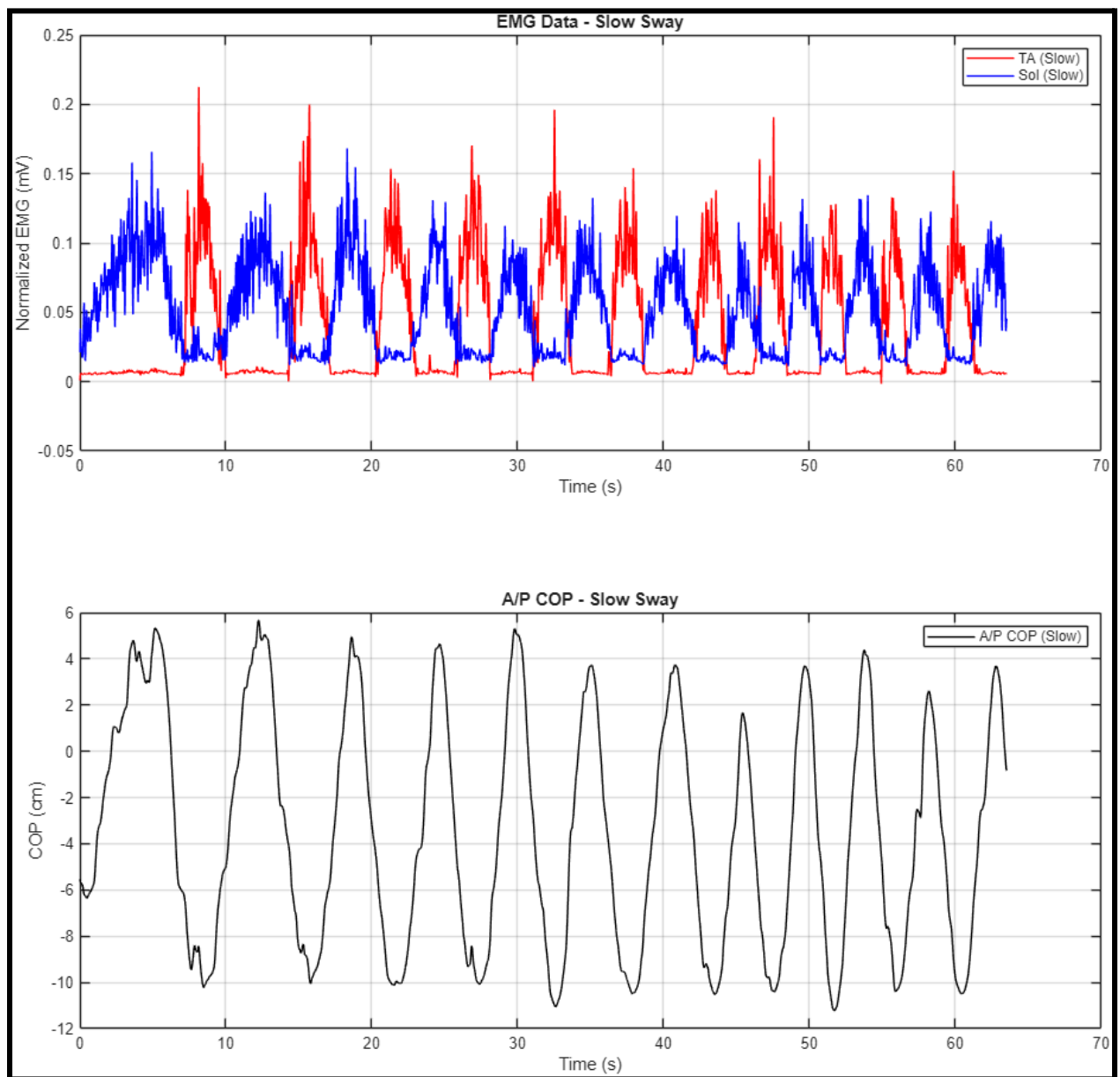


Figure 9: EMG and COP Comparison Plot for Slow Sway [A.2]

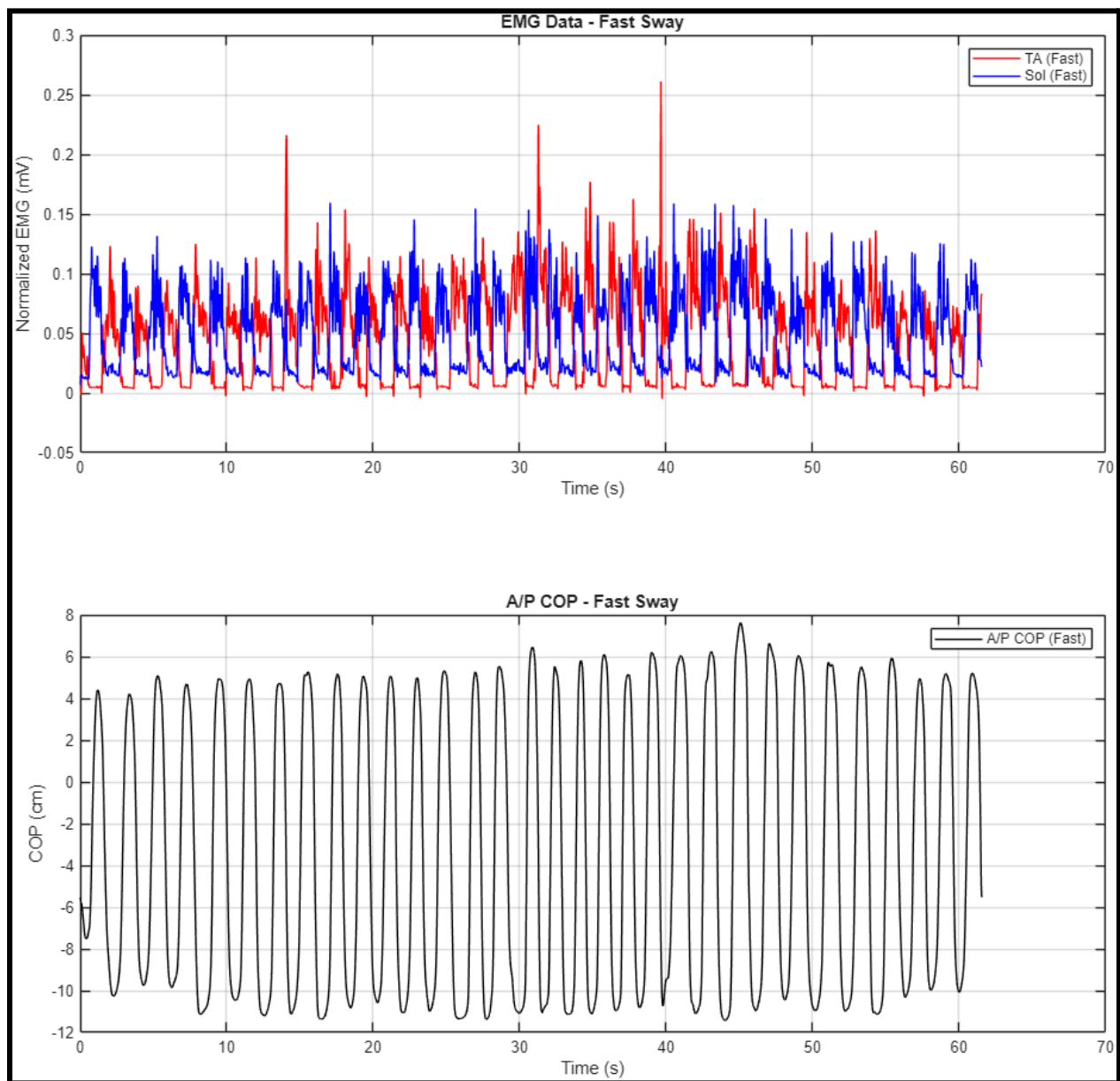


Figure 10: EMG and COP Comparison for Fast Sway [A.2]

2.4: Discuss all notable differences between the muscle reaction in slow and fast sway, comment on how soleus and tibialis muscles work together based on their EMG responses.

The most notable difference between the muscle reactions can be observed in the EMG recordings in Figures 9 and 10. In the slow sway (Figure 9), there is a clearer separation between the activation of the soleus and tibialis anterior muscles, indicating that these muscles alternate their activity more distinctly as the body moves between anterior and posterior sway. This pattern suggests that during slow sway, the muscles have time to activate and relax independently, maintaining balance with less simultaneous engagement.

Conversely, in the fast sway (Figure 10), there is less separation between muscle contractions, indicating more

co-contraction of the soleus and tibialis anterior. This co-contraction is necessary for stabilizing the body quickly and maintaining balance during rapid movement. The simultaneous activation of these muscles allows for better control and force generation, which is crucial for maintaining stability at a faster pace. This synergy between the muscles shows how they work together to achieve the rapid and coordinated response needed for fast sway movements.

Discussion

Discuss how Part 1 results are similar to Lab 2

Part 1 evaluated postural stability by contrasting eyes open and eyes closed conditions using stabilograms (Figures 1,2) and quantitative measures total excursion (TOTEX) and mean velocity (MVELO) to visualize center of pressure (COP) paths and quantify sway magnitude and speed. This combined approach provided both a depiction of COP paths and numerical indices that captured differences in sway behavior across visual conditions.

In Lab 2, stability was compared across datasets using stabilograms in conjunction with (RMSE) as the primary metric. While RMSE is useful for summarizing deviation from a reference, it conveys less about temporal and directional characteristics of movement than TOTEX or MVELO, which more directly reflect path length and sway dynamics.

Across both labs, graphical representation along with numerical metrics improved the precision and comparability of stability assessments. The integration of visual COP patterns with summary measures (path length metrics and RMSE) yielded a more comprehensive picture of balance control and facilitated clearer condition comparisons.

Comment on differences in mean velocity and TOTEX

As mentioned in Part 1.4 and supported by the results, the mean velocity and TOTEX metrics for the eyes open (EO) condition are smaller and suggest greater stability compared to the eyes closed (EC) condition. The excursion (TOTEX) measures the total distance traveled by the center of pressure (COP), providing insight into the overall movement and sway of the subject. In the EC condition, the higher TOTEX values indicate that the subject's body swayed more significantly, covering a larger distance as it moved to maintain balance. This greater movement suggests a reduced ability to stabilize without visual input.

Similarly, mean velocity measures the average speed of the COP movement, and the higher mean velocities in the EC condition imply that not only does the COP travel further, but it does so at a faster

rate. This combination of increased TOTEX and mean velocity highlights the greater effort required to maintain balance without visual cues, as the body must make quicker, more frequent adjustments. Overall, these metrics confirm that visual input is essential for minimizing postural sway and maintaining stability.

What are the differences seen in fast vs. slow sway?

Referring to Part 2.4, the key distinction between fast and slow sway is the EMG activation pattern of the soleus and tibialis anterior. During slow sway, these exhibit clearer alternation, with out of phase bursts that allow each muscle to activate and relax independently as the body shifts anterior-posterior, yielding smoother, finely modulated balance corrections.

In contrast, fast sway shows reduced separation and greater co-contraction of the two muscles, a strategy that increases ankle joint stiffness to rapidly stabilize the body and corrective torques must rise quickly. This simultaneous activation facilitates swift, coordinated responses under time critical conditions, supporting stability during rapid movements.

Discuss how soleus and tibialis muscles work together based on their EMG responses

According to Part 2.4, the soleus and tibialis anterior exhibit task dependent coordination that changes with sway speed. During slow sway, their activations are more out of phase, with one muscle activating as the other relaxes, enabling smoother ankle torques and balance corrections with minimal simultaneous engagement.

In fast sway, the activation patterns overlap more, showing increased co-contraction that stiffens the ankle for rapid stabilization. This simultaneous activation enhances immediate torque availability and control, which is crucial for maintaining balance under time critical conditions.

Together, the transition from alternating activity in slow sway to co-contraction in fast sway highlights adaptive synergy. The system trades efficiency for quicker response as task demands, optimizing stability across different movement speeds.

Conclusion

In conclusion, this lab provided valuable insights on how muscle activation and sway control contributes to balance and stability, which are essential for rehabilitation and physical therapy. From analyzing center of pressure (COP) and sway velocity data in both eyes open and eyes closed conditions it was evident of increased sway when visual cues were removed. EMG data analysis of the soleus and tibialis anterior during fast and slow sway further

demonstrated how these muscles coordinate though independent activation or cocontraction depending on the sway speed as both are essential for maintaining balance. With this in mind, design considerations for Functional Electrical Stimulation (FES) in rehabilitation can be strategically activating specific muscles in response to detected sway, ultimately supporting targeted and adaptive balance improvements. This lab emphasized the importance of integrating COP and muscle activation analysis when developing personalized rehabilitation protocols in order to enhance balance, stability, and mobility for those with muscular and neurological challenges.

References

- [1] Y. Ivanenko and V. S. Gurfinkel, "Human Postural Control," *Frontiers in Neuroscience*, vol. 12, no. 171, Mar. 2018, doi: <https://doi.org/10.3389/fnins.2018.00171>.
- [2] A. Ruhe, R. Fejer, and B. Walker, "Center of pressure excursion as a measure of balance performance in patients with non-specific low back pain compared to healthy controls: a systematic review of the literature," *European Spine Journal*, vol. 20, no. 3, pp. 358-368, Aug. 2010, doi: <https://doi.org/10.1007/s00586-010-1543-2>.
- [3] D. Gurve & , L. Minkowski , and D. Santillo, "Lab 3: Center of pressure (COP) and sway analysis," Sep. 2025.
- [4] R. Ritzmann, K. Freyler, A. Werkhausen, and A. Gollhofer, "Changes in Balance Strategy and Neuromuscular Control during a Fatiguing Balance Task—A Study in Perturbed Unilateral Stance," *Frontiers in Human Neuroscience*, vol. 10, Jun. 2016, doi: <https://doi.org/10.3389/fnhum.2016.00289>.
- [5] M. K. Dummar et al., "The Convergent Validity of the SWAY Balance Application to Assess Postural Stability in Military Cadets Recovering from Concussion," *International Journal of Sports Physical Therapy*, vol. 19, no. 2, Feb. 2024, doi: <https://doi.org/10.26603/001c.92234>.
- [6] [1]J. A. Robertson, J. J. Eng, and C. Hung, "The Effect of Functional Electrical Stimulation on Balance Function and Balance Confidence in Community-Dwelling Individuals with Stroke," *Physiotherapy Canada*, vol. 62, no. 2, pp. 114-119, Apr. 2010, doi: <https://doi.org/10.3138/physio.62.2.114>.

- [7] D. A. Winter, A. E. Patla, M. Ishac, and W. H. Gage, "Motor mechanisms of balance during quiet standing," J Electromyogr Kinesiol, vol. 13, pp. 49-56, 2003

BME705: Rehabilitation Engineering

Lab 3: Centre of Pressure & Sway Analysis

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%  
% BME705: Rehabilitation Engineering  
% Lab 3: Center of Pressure (COP) and Sway Analysis  
%  
% Created by: Devon Santillo and Dharmendra Gurve, 2020  
% Updated by: Dan Genkin, 2021  
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%  
% Name: Tanvir Hassan, Ilina Petrova  
% Student ID: 501104056, 501126967
```

Loading Le Files

```
clear; clc; close all;  
% Load the data and define the sampling frequency  
fs = 2000;  
load("ECEO_data4.mat");  
load("sway_data3.mat")
```

PART 1

```
% analyzeFP function: "Given" (do not edit): convert raw forceplate data  
% INPUT = 16 channels of raw forceplate data;  
% OUTPUT: xCOPL, yCOPL, xCOPR, yCOPR, Fz_R, Fz_L  
% Note: Rv_R and Rv_L are vertical forces of right and left forceplate  
  
% You can analyze the eyes open (EO) and eyes closed (EC) data separately in a  
% repeated process or in a logic loop  
% but remember you cant have matrices as the iterative variable in a loop
```

1.1 Calcualte and Plot the COPnet of the eyes open data

```
% Apply the provided function to obtain necessary outputs for further analysis  
[xCOPL, yCOPL, xCOPR, yCOPR, Rv_R, Fz_L] = analyzeFP(qsEO_fp);  
  
% Shift all X-coordinates to the middle of the split force plate by +/- 12.5825cm  
xCOPL = xCOPL - 12.5825;  
xCOPR = xCOPR + 12.5825;  
  
% Filter all data  
% Use of a Low-pass filter: a Butterworth is recommended  
% define corner frequency and normalized cut-off frequency  
fc=10;  
Wn = fc / (fs / 2);  
[b, a] = butter(4, Wn);  
xCOPL = filtfilt(b, a, xCOPL);  
yCOPL = filtfilt(b, a, yCOPL);
```

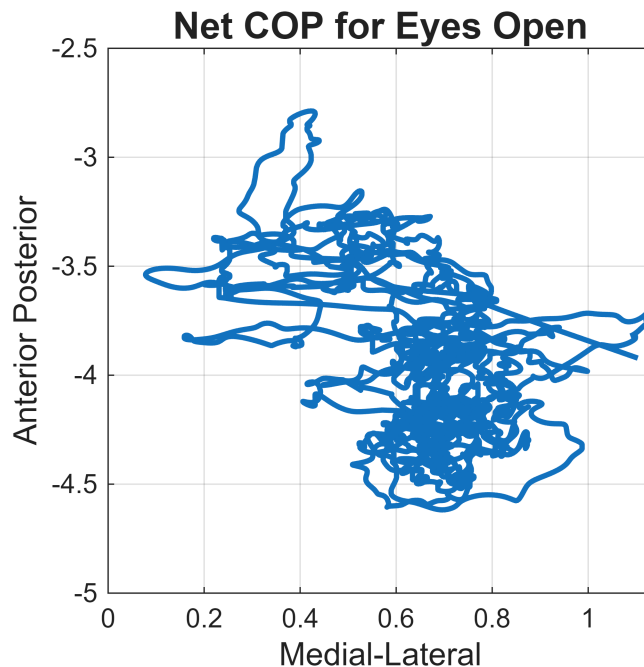
```

xCOPR = filtfilt(b, a, xCOPR);
yCOPR = filtfilt(b, a, yCOPR);
Rv_R = filtfilt(b, a, Rv_R);
Fz_L = filtfilt(b, a, Fz_L);

% Calculate COPnet for overall system using Equation A from the lab manual (Winter
et al., 2003 [1]):
xCOP_net = (xCOP_L .* (Fz_L ./ (Fz_L + Rv_R))) + (xCOPR .* (Rv_R ./ (Fz_L + Rv_R)));
yCOP_net = (yCOP_L .* (Fz_L ./ (Fz_L + Rv_R))) + (yCOPR .* (Rv_R ./ (Fz_L + Rv_R)));

% Plot net COP (the stabilograms)
figure;
plot(xCOP_net, yCOP_net, 'LineWidth', 2);
title("Net COP for Eyes Open", 'FontSize', 14);
xlabel("Medial-Lateral", 'FontSize', 12);
ylabel("Anterior Posterior", 'FontSize', 12);
grid on;
axis equal;
axis square; % Make the plot square

```



1.2 Calcualte and plot the COPnet of the eyes closed data

```

[xCOPL2, yCOPL2, xCOPR2, yCOPR2, Rv_R2, Fz_L2] = analyzeFP(qsEC_fp);
% Shift all X-coordinates to the middle of the split force plate by +/- 12.5825 cm
xCOPL2 = xCOPL2 - 12.5825;
xCOPR2 = xCOPR2 + 12.5825;

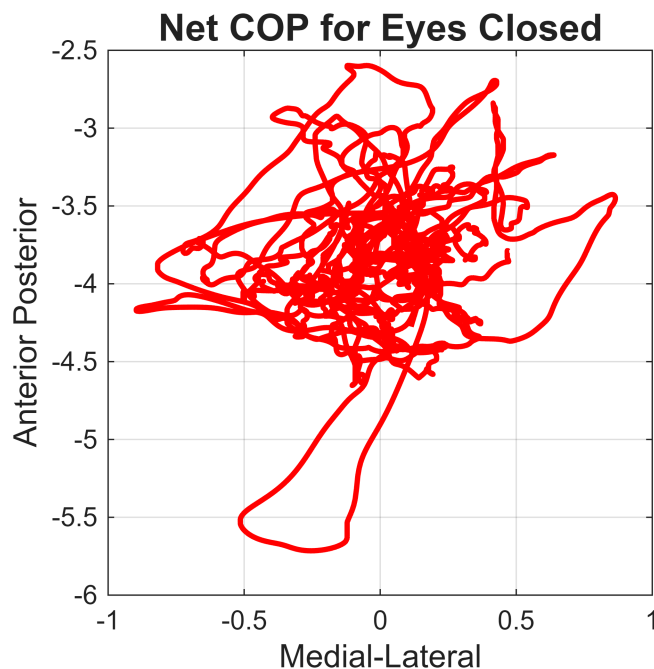
% Filter all data
% Use of a Low-pass filter: a Butterworth is recommended

```

```

% Define corner frequency and normalized cut-off frequency
fc = 10;
Wn = fc / (fs / 2);
[b, a] = butter(4, Wn);
xCOP_L2 = filtfilt(b, a, xCOP_L2);
yCOP_L2 = filtfilt(b, a, yCOP_L2);
xCOP_R2 = filtfilt(b, a, xCOP_R2);
yCOP_R2 = filtfilt(b, a, yCOP_R2);
Rv_R2 = filtfilt(b, a, Rv_R2);
Fz_L2 = filtfilt(b, a, Fz_L2);
% Calculate COPnet for overall system using Equation A from the lab manual (Winter
et al., 2003 [1]):
xCOP_net2 = (xCOP_L2 .* (Fz_L2 ./ (Fz_L2 + Rv_R2))) + (xCOP_R2 .* (Rv_R2 ./ (Fz_L2 +
Rv_R2)));
yCOP_net2 = (yCOP_L2 .* (Fz_L2 ./ (Fz_L2 + Rv_R2))) + (yCOP_R2 .* (Rv_R2 ./ (Fz_L2 +
Rv_R2)));
% Plot net COP (the stabilograms)
figure;
plot(xCOP_net2, yCOP_net2, 'LineWidth', 2, 'Color', 'r');
title("Net COP for Eyes Closed", 'FontSize', 14);
xlabel("Medial-Lateral", 'FontSize', 12);
ylabel("Anterior Posterior", 'FontSize', 12);
grid on;
axis equal;
axis square; % Make the plot square

```



1.3 Find total excursions and mean velocity using equations B-I

```

% Using the method described in Prieto et al., 1994 paper [2]
% Select 20 seconds of the data in the time frame from 40-60 seconds
T=20;

```

```

start_point = fs*40;
end_point = fs*60;
xCOP_segment = xCOP_net(start_point:end_point);
yCOP_segment = yCOP_net(start_point:end_point);

% Downsample A/P and M/L axes to 100 Hz
fs_new = 100;
xCOP_downsampled = downsample(xCOP_segment, fs / fs_new);
yCOP_downsampled = downsample(yCOP_segment, fs / fs_new);

% Zero-mean AP and ML : Equations B & C (Prieto et al., 1994)
xCOP_zero_mean = xCOP_downsampled - mean(xCOP_downsampled); %AP
yCOP_zero_mean = yCOP_downsampled - mean(yCOP_downsampled); %ML

% Total Excursion: Equations D-F (Prieto et al., 1994)
% Calculate TOTEX ML, AP, Net for Open and Closed
TOTEXap = 0;
TOTEXml = 0;
TOTEX = 0;
for i = 1:length(xCOP_zero_mean) - 1
    TOTEXap = TOTEXap + abs(xCOP_zero_mean(i+1) - xCOP_zero_mean(i));
    TOTEXml = TOTEXml + abs(yCOP_zero_mean(i+1) - yCOP_zero_mean(i));
    TOTEX = TOTEX + sqrt((xCOP_zero_mean(i+1) - xCOP_zero_mean(i))^2 +
(yCOP_zero_mean(i+1) - yCOP_zero_mean(i))^2);
end

% Mean velocity: Equations G-I (Prieto et al., 1994)
mean_velocity_ap = TOTEXap / T;
mean_velocity_ml = TOTEXml / T;
mean_velocity_net = TOTEX / T;

```

For Eyes Closed

```

xCOP_segment2 = xCOP_net2(start_point:end_point);
yCOP_segment2 = yCOP_net2(start_point:end_point);

% Downsample A/P and M/L axes to 100 Hz
xCOP_downsampled2 = downsample(xCOP_segment2, fs / fs_new);
yCOP_downsampled2 = downsample(yCOP_segment2, fs / fs_new);

% Zero-mean AP and ML : Equations B & C (Prieto et al., 1994)
xCOP_zero_mean2 = xCOP_downsampled2 - mean(xCOP_downsampled2); % AP
yCOP_zero_mean2 = yCOP_downsampled2 - mean(yCOP_downsampled2); % ML

% Total Excursion: Equations D-F (Prieto et al., 1994)
% Calculate TOTEX ML, AP, Net for Eyes Closed
TOTEXap2 = 0;
TOTEXml2 = 0;

```

```

TOTEX2 = 0;
for i = 1:length(xCOP_zero_mean2) - 1
    TOTEXap2 = TOTEXap2 + abs(xCOP_zero_mean2(i+1) - xCOP_zero_mean2(i));
    TOTEXml2 = TOTEXml2 + abs(yCOP_zero_mean2(i+1) - yCOP_zero_mean2(i));
    TOTEX2 = TOTEX2 + sqrt((xCOP_zero_mean2(i+1) - xCOP_zero_mean2(i))^2 +
(yCOP_zero_mean2(i+1) - yCOP_zero_mean2(i))^2);
end

% Mean velocity: Equations G-I (Prieto et al., 1994)
mean_velocity_ap2 = TOTEXap2 / T;
mean_velocity_ml2 = TOTEXml2 / T;
mean_velocity_net2 = TOTEX2 / T;
% Display results in a formatted table for lab report
Metrics = {'Total Excursion (AP) (cm)'; 'Total Excursion (ML) (cm)'; 'Total
Excursion (Net) (cm)'; ...
'Mean Velocity (AP) (cm/s)'; 'Mean Velocity (ML) (cm/s)'; 'Mean Velocity
(Net) (cm/s)'};
EyesOpen = [TOTEXap; TOTEXml; TOTEX; mean_velocity_ap; mean_velocity_ml;
mean_velocity_net];
EyesClosed = [TOTEXap2; TOTEXml2; TOTEX2; mean_velocity_ap2; mean_velocity_ml2;
mean_velocity_net2];
resultsTable = table(Metrics, EyesOpen, EyesClosed);

% Format the table for better readability in lab report
disp('Results Summary:');

```

Results Summary:

```
disp(resultsTable);
```

Metrics	EyesOpen	EyesClosed
{'Total Excursion (AP) (cm)' }	7.1619	8.8899
{'Total Excursion (ML) (cm)' }	9.6363	12.404
{'Total Excursion (Net) (cm)'} }	13.182	16.632
{'Mean Velocity (AP) (cm/s)' }	0.35809	0.44449
{'Mean Velocity (ML) (cm/s)' }	0.48181	0.62022
{'Mean Velocity (Net) (cm/s)'} }	0.65909	0.83162

PART 2

2.1 Processing EMG data: Consider the timing of activation of the Sol. and TA muscles with respect to the COP sway in the AP direction.

```

G = 1000; %gain
fs = 2000; % sampling rate 200Hz

% Define time vectors
t_fast = (0:length(fp_fast)-1) / fs;
t_slow = (0:length(fp_slow)-1) / fs;

% Process sol_fast

```

```

sol_fast = sol_fast / G; % Normalize gain
sol_fast = sol_fast * 1000; % Convert to mV
sol_fast_rectified = abs(sol_fast);
sol_fast_rectified_max = max(sol_fast_rectified);
sol_fast_normalized = sol_fast_rectified / sol_fast_rectified_max;
sol_fast_filtered = filtfilt(b, a, sol_fast_normalized); % Apply zero-phase
filtering

% Process sol_slow
sol_slow = sol_slow / G; % Normalize gain
sol_slow = sol_slow * 1000; % Convert to mV
sol_slow_rectified = abs(sol_slow);
sol_slow_rectified_max = max(sol_slow_rectified);
sol_slow_normalized = sol_slow_rectified / sol_slow_rectified_max;
sol_slow_filtered = filtfilt(b, a, sol_slow_normalized); % Apply zero-phase
filtering

% Process ta_fast
ta_fast = ta_fast / G; % Normalize gain
ta_fast = ta_fast * 1000; % Convert to mV
ta_fast_rectified = abs(ta_fast);
ta_fast_rectified_max = max(ta_fast_rectified);
ta_fast_normalized = ta_fast_rectified / ta_fast_rectified_max;
ta_fast_filtered = filtfilt(b, a, ta_fast_normalized); % Apply zero-phase filtering

% Process ta_slow
ta_slow = ta_slow / G; % Normalize gain
ta_slow = ta_slow * 1000; % Convert to mV
ta_slow_rectified = abs(ta_slow);
ta_slow_rectified_max = max(ta_slow_rectified);
ta_slow_normalized = ta_slow_rectified / ta_slow_rectified_max;
ta_slow_filtered = filtfilt(b, a, ta_slow_normalized); % Apply zero-phase filtering

function processeEMG(EMGdata, t_EMG, EMG_rectified, EMG_rectified_max,
EMG_normalized, EMG_filtered)
    fs = 2000; % sampling rate 200Hz
    % Original EMG plot
    figure;
    subplot(4,1,1);
    plot(t_EMG, EMGdata, 'LineWidth', 1.5);
    title(['Original EMG (' inputname(1) ')'], 'FontSize', 14);
    xlabel('Time (s)', 'FontSize', 12);
    ylabel('EMG (mV)', 'FontSize', 12);
    grid on;
    set(gca, 'FontSize', 12);

    % Rectification
    EMG_rectified = abs(EMGdata);
    subplot(4,1,2);
    plot(t_EMG, EMG_rectified, 'LineWidth', 1.5);

```



```

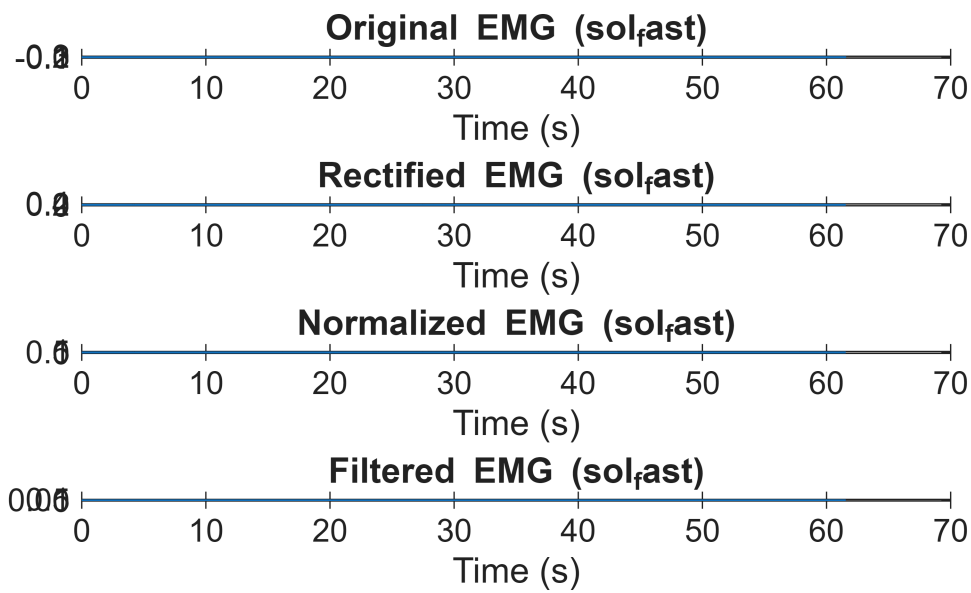
title(['Rectified EMG (' inputname(1) ')'], 'FontSize', 14);
xlabel('Time (s)', 'FontSize', 12);
ylabel('Rectified EMG (mV)', 'FontSize', 12);
grid on;
set(gca, 'FontSize', 12);

% Normalization
EMG_rectified_max = max(EMG_rectified);
EMG_normalized = EMG_rectified / EMG_rectified_max;
subplot(4,1,3);
plot(t_EMG, EMG_normalized, 'LineWidth', 1.5);
title(['Normalized EMG (' inputname(1) ')'], 'FontSize', 14);
xlabel('Time (s)', 'FontSize', 12);
ylabel('Normalized EMG (mV)', 'FontSize', 12);
grid on;
set(gca, 'FontSize', 12);

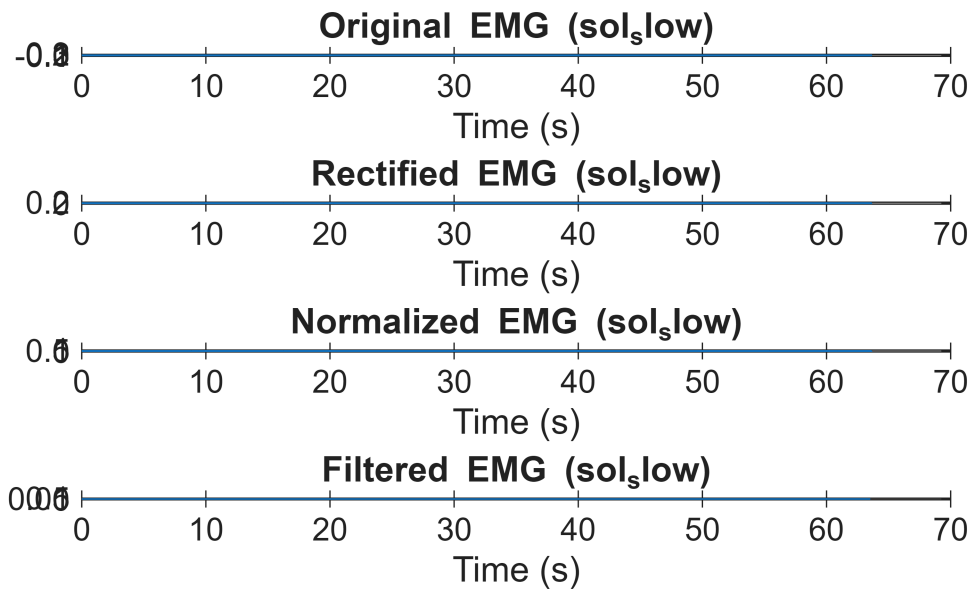
% Signal Envelope Filtering (2.5 Hz low-pass filter)
fc = 2.5; % cutoff frequency
[b, a] = butter(4, fc/(fs/2));
EMG_filtered = filtfilt(b, a, EMG_normalized); % Use filtfilt for zero-phase
distortion
subplot(4,1,4);
plot(t_EMG, EMG_filtered, 'LineWidth', 1.5);
title(['Filtered EMG (' inputname(1) ')'], 'FontSize', 14);
xlabel('Time (s)', 'FontSize', 12);
ylabel('Filtered EMG (mV)', 'FontSize', 12);
grid on;
set(gca, 'FontSize', 12);
end

% Call the function with sol_fast and t_fast
processEMG(sol_fast, t_fast, sol_fast_rectified, sol_fast_rectified_max,
sol_fast_normalized, sol_fast_filtered);

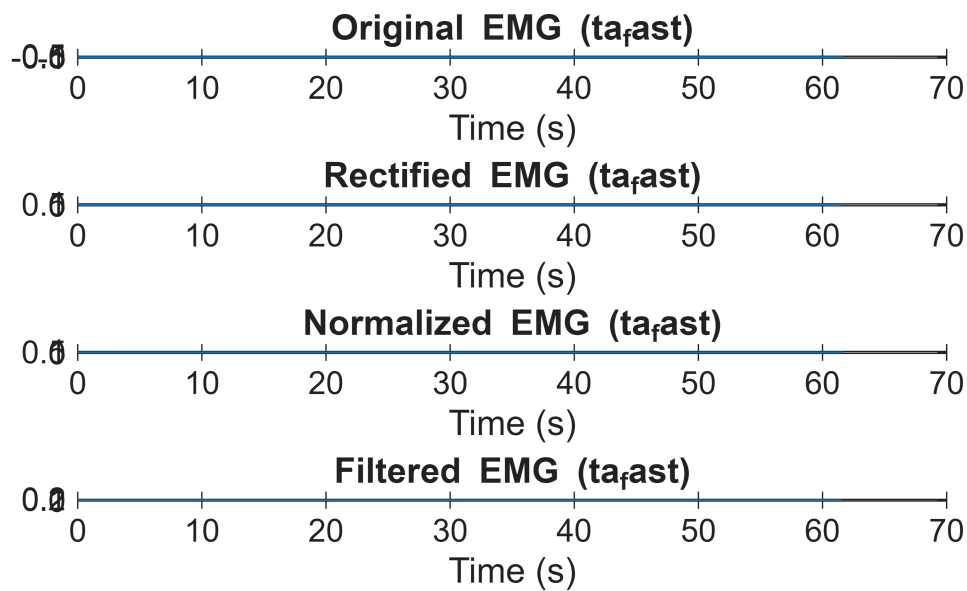
```



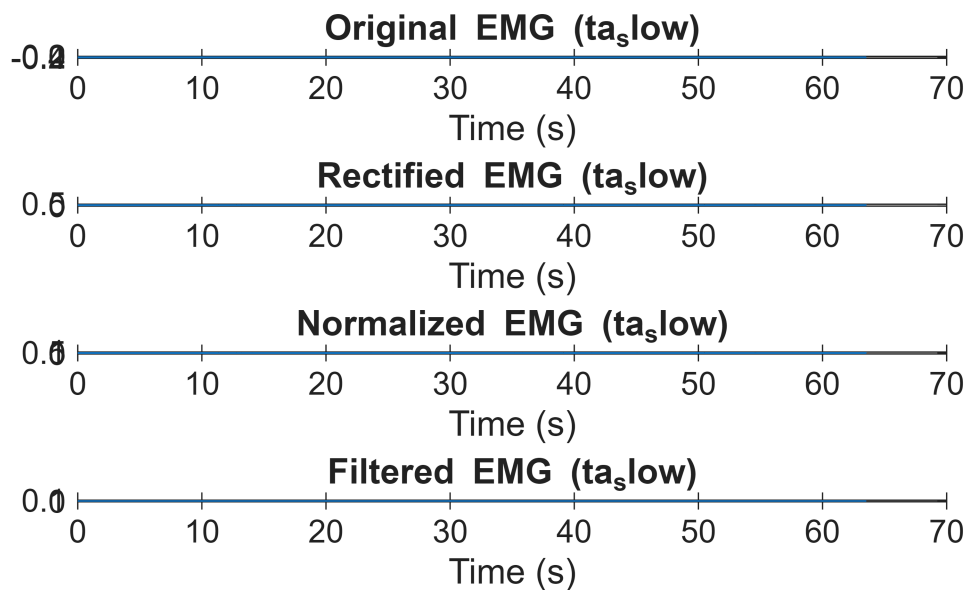
```
processEMG(sol_slow, t_slow, sol_slow_rectified, sol_slow_rectified_max,
sol_slow_normalized, sol_slow_filtered);
```



```
processEMG(ta_fast, t_fast, ta_fast_rectified, ta_fast_rectified_max,
ta_fast_normalized, ta_fast_filtered);
```



```
processEMG(ta_slow, t_slow, ta_slow_rectified, ta_slow_rectified_max,
ta_slow_normalized, ta_slow_filtered);
```



2.2 Processing sway data (ML, AP) - Repeat force plate analysis

```
% This section should follow the first part of the analysis
% call the analyzeFP function again (do not edit it)
[xCOPL, yCOPL, xCOPR, yCOPR, Rv_R, Fz_L] = analyzeFP(fp_fast);
[xCOPL2, yCOPL2, xCOPR2, yCOPR2, Rv_R2, Fz_L2] = analyzeFP(fp_slow);
% Apply the provided function to obtain necessary outputs for further analysis
```

```

% Shift X-coordinates to the middle of the split force plate by +/- 12.5825cm
xCOPL = xCOPL - 12.5825;
xCOPR = xCOPR + 12.5825;
xCOPL2 = xCOPL2 - 12.5825;
xCOPR2 = xCOPR2 + 12.5825;

% Filter all the slow & fast, L & R COP data
fc = 10;
Wn = fc / (fs / 2);
[b, a] = butter(4, Wn);

% Apply filtering to fast sway data
xCOPL = filtfilt(b, a, xCOPL);
yCOPL = filtfilt(b, a, yCOPL);
xCOPR = filtfilt(b, a, xCOPR);
yCOPR = filtfilt(b, a, yCOPR);
Rv_R = filtfilt(b, a, Rv_R);
Fz_L = filtfilt(b, a, Fz_L);

% Apply filtering to slow sway data
xCOPL2 = filtfilt(b, a, xCOPL2);
yCOPL2 = filtfilt(b, a, yCOPL2);
xCOPR2 = filtfilt(b, a, xCOPR2);
yCOPR2 = filtfilt(b, a, yCOPR2);
Rv_R2 = filtfilt(b, a, Rv_R2);
Fz_L2 = filtfilt(b, a, Fz_L2);

% Calculate COPnet for overall system of both fast and slow sway using Equation A
from the lab manual
% Fast sway
xCOP_net = (xCOPL .* (Fz_L ./ (Fz_L + Rv_R))) + (xCOPR .* (Rv_R ./ (Fz_L + Rv_R)));
yCOP_net = (yCOPL .* (Fz_L ./ (Fz_L + Rv_R))) + (yCOPR .* (Rv_R ./ (Fz_L + Rv_R)));
% Slow sway
xCOP_net2 = (xCOPL2 .* (Fz_L2 ./ (Fz_L2 + Rv_R2))) + (xCOPR2 .* (Rv_R2 ./ (Fz_L2 +
Rv_R2)));
yCOP_net2 = (yCOPL2 .* (Fz_L2 ./ (Fz_L2 + Rv_R2))) + (yCOPR2 .* (Rv_R2 ./ (Fz_L2 +
Rv_R2)));

% Plot Fast and slow COPnet Stabilograms with improved visualization
figure;

% Fast sway
subplot(2,1,1);
plot(xCOP_net, yCOP_net, 'LineWidth', 2, 'Color', 'b'); % Increased line width and
changed color
title('COPnet Stabilogram - Fast Sway', 'FontSize', 14);
xlabel('Medial-Lateral (cm)', 'FontSize', 12);
ylabel('Anterior-Posterior (cm)', 'FontSize', 12);
grid on;
axis equal;

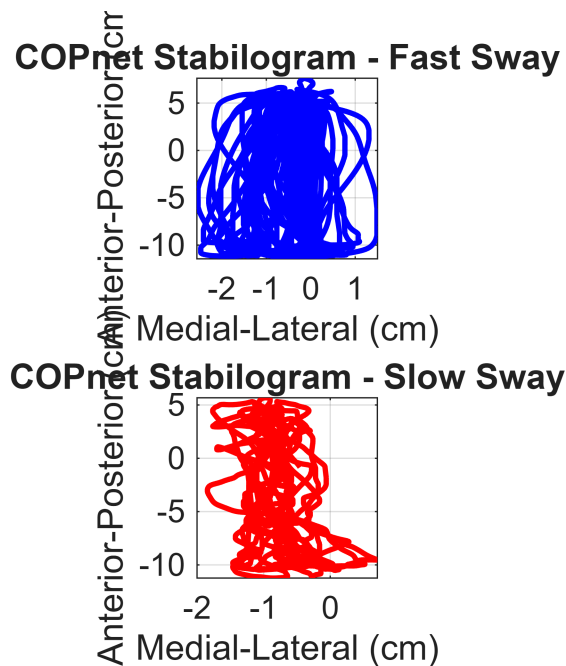
```

```

axis square; % Make the plot square
set(gca, 'FontSize', 12);

% Slow sway
subplot(2,1,2);
plot(xCOP_net2, yCOP_net2, 'LineWidth', 2, 'Color', 'r'); % Increased line width
and changed color
title('COPnet Stabilogram - Slow Sway', 'FontSize', 14);
xlabel('Medial-Lateral (cm)', 'FontSize', 12);
ylabel('Anterior-Posterior (cm)', 'FontSize', 12);
grid on;
axis equal;
axis square; % Make the plot square
set(gca, 'FontSize', 12);

```



2.3 Plot the slow and fast EMG and the Sway data together

```

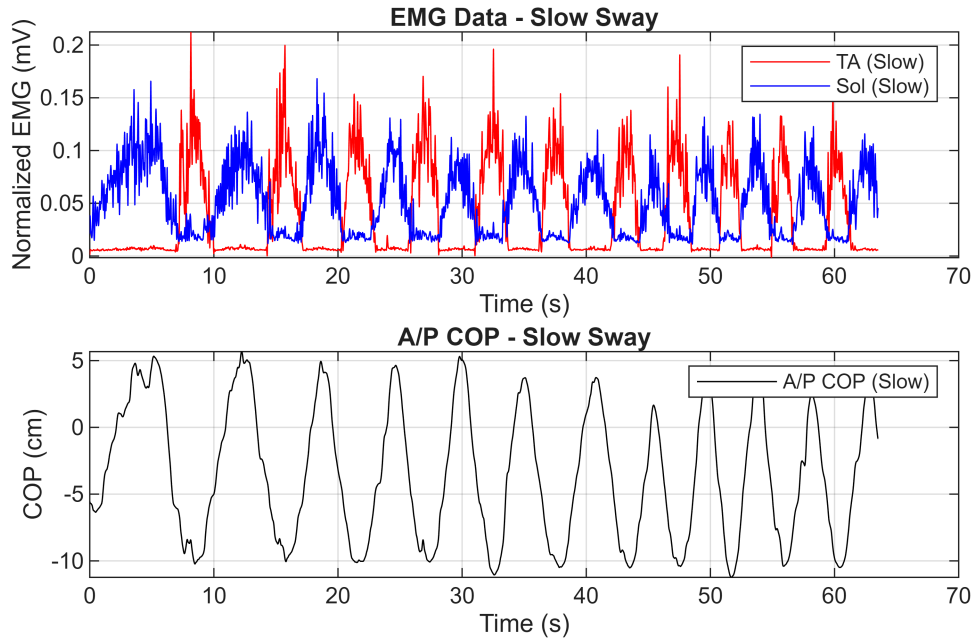
% slow
figure;
% Subplot 1: EMG data for slow sway (TA and Sol)
subplot(2,1,1);
plot(t_slow, ta_slow_filtered, 'r', 'DisplayName', 'TA (Slow)');
hold on;
plot(t_slow, sol_slow_filtered, 'b', 'DisplayName', 'Sol (Slow)');
title('EMG Data - Slow Sway');
xlabel('Time (s)');
ylabel('Normalized EMG (mV)');
legend('show');
grid;
hold off;
% Subplot 2: A/P COP for slow sway

```

```

subplot(2,1,2);
plot(t_slow, yCOP_net2, 'k', 'DisplayName', 'A/P COP (Slow)');
title('A/P COP - Slow Sway');
xlabel('Time (s)');
ylabel('COP (cm)');
grid;
legend('show');

```



```

% fast
figure;
% Subplot 1: EMG data for fast sway (TA and Sol)
subplot(2,1,1);
plot(t_fast, ta_fast_filtered, 'r', 'DisplayName', 'TA (Fast)');
hold on;
plot(t_fast, sol_fast_filtered, 'b', 'DisplayName', 'Sol (Fast)');
title('EMG Data - Fast Sway');
xlabel('Time (s)');
ylabel('Normalized EMG (mV)');
legend('show');
grid;
hold off;
% Subplot 2: A/P COP for fast sway
subplot(2,1,2);
plot(t_fast, yCOP_net, 'k', 'DisplayName', 'A/P COP (Fast)');
title('A/P COP - Fast Sway');
xlabel('Time (s)');
ylabel('COP (cm)');
grid;
legend('show');

```

